R13

Code No: 117EE

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is a Shell?	[2]
b)	Explain briefly about Sed-Scripts.	[3]
c)	What is a File?	[2]
d)	What is a Hard Link?	[3]
e)	Explain briefly about Abort Function.	[2]
f)	What is a Process?	[3]
g)	Define IPC.	[2]
h)	Discuss briefly about "popen" library function.	[3]
i)	What is Shared Memory?	[2]
j)	Explain briefly about "setsockopt" system call.	[3]

PART – B

(50 Marks)

2.a)	List out File Handling Utilities.	
b)	Discuss in detail about System commands in awk.	[5+5]
	OR	
3.a)	Enumerate Text processing utilities and Backup utilities.	
b)	Explain in detail about various control structures in Shell Programming.	[5+5]
4.a)	Write a detailed note on Kernel Support for Files.	
b)	What is a Directory? Explain in detail about Scanning Directories.	[5+5]
	OR	
5.a)	Explain in detail about File Permissions and File Ownership commands.	
b)	How do we obtain Current Working Directory? Explain Directory Contents.	[5+5]

6.a) Discuss Layout of a C program image in Main Memory.
b) Explain in detail about Signal generation and handling. [5+5]

OR

7.a) Explain in detail about Kernel support for Process.
b) Differentiate between Reliable and Unreliable Signals. [5+5]

8. What is a Pipe? Explain in detail about differences between unnamed and named Pipes. [10]

OR

9. What is a Semaphore? Explain in detail about APIs for Semaphores. [10]

10.a) Discuss in detail about Kernel support for Shared Memory.
b) Explain in detail about Single Server-Client connection with an example. [5+5]

OR

11.a) Discuss about IPC over a Network.
b) Explain in detail about Berkeley Sockets. [5+5]

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